

Multi-objective optimization and numerical modelling of helical coil spring for automotive application

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ABSTRACT

In this paper, a helical compression spring used in commercial transport three-wheeler has been modelled as a multi-objective optimisation problem. The objective functions considered in this study are reducing the volume of the spring and increasing the strain energy of the spring used in the light automotive. Minimizing the volume of the spring leads to reducing the cost of the spring. The utility and functionality of the spring can be improved by increasing the strain energy of the helical spring. Design parameters such as diameter of the spring wire, diameter of the coil spring and number of active turns of the spring are optimized in this study by using multi-objective optimisation formulation. Heuristic techniques such as Genetic Algorithms are used to find Pareto optimal solutions for a wide variety of engineering problems with multiple objectives. In this work, non-dominated solutions are generated using Non-Dominated Sorting GA-II approach. The Pareto-optimal solutions obtained in this study are validated using a numerical approach using ANSYS workbench.

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1. Introduction

The industrial revolution has set into motion a wide variety of technological advancements in the field of Engineering. Product designs are expected to be optimum from the manufacturer's and as well as the customer's viewpoint. For meeting the demand of the customer and to survive in the competitive environment, manufacturers are forced to optimize their products by integrating optimisation approaches in their design process. In the field of engineering design, optimisation has been one of the focus areas of the designers for improving the design in terms of quality, cost and time.

The optimisation involves finding the minimum or maximum value of a function under given circumstances. The optimisation problem may be classified into various categories such as a) linear b) non-linear c) optimal-control d) quadratic e) geometric programming, f) Integer programming, g) stochastic programming h) separable programming and i) multi-objective programming. Based on the nature of decision variables, the optimisation problems are classified to a discrete and continuous optimisation problem. Numerous techniques are available in the literature to solve

optimisation problems. For different problem formulations, different optimisation techniques are used to arrive optimal or near-optimal solutions. The classical method of optimisation techniques is used to solve continuous functions with derivative information. Often classical methods are trapped in local optima. Whereas direct methods of optimisation work with function values arrived at every iteration. Indirect methods, tackling convergence of algorithms is an important issue.

In general, most of the real-world optimisation problems are non-linear, and design variables must be satisfied with multiple objectives. For simplicity, real-world non-linear problems are often converted into linear problems, and multi-objective problems are solved as a single objective optimisation problem. Heuristic approaches are computationally intensive and are designed to find optimal or near-optimal solutions to the problem. Due to advancement in computing technology, heuristic approaches are being used by the researches and industry practitioners for getting solutions to complex discrete and continuous optimisation problems with lesser computational complexity.

1.1. Meta-Heuristic techniques

Heuristic algorithms are developed to reduce the time complexity in solving hard and large scale optimisation problems. Heuristic

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Techniques are proved to be very powerful and most sought out methods for solving engineering optimisation problems in recent times. The ability of the heuristics mainly depends on exploration and exploitation capability to identify the quality solution in the design space. Heuristics which combines the exploitation and exploration principles intelligently to find optimal or near-optimal solutions are called metaheuristic algorithms.

Some of the most promising *meta*-heuristic methods for solving optimization problems include, Simulated-Annealing (SA) [1], Tabu-Search [2], Evolutionary-Algorithms (EA) like Genetic-Algorithms (GA) [3], Ant-Colony-Optimization [4], Differential Evolution (DE) [5] and Particle Swarm Optimization Algorithm (PSO) [6]. For Design optimisation problems, Heuristics are being used by the researchers. Sandgren [7] solved design optimisation problems using branch and bound method for a variety of engineering optimisation problems. Optimal design of welded beams was studied by Deb [8]. Deb and Goyal [9] optimised engineering designs using genetic search procedures. Cao and Wu [10] proposed an evolutionary programming approach for optimizing engineering design for mechanical components. He et al., [11] proposed PSO algorithm for design optimisation. The challenges and future directions of design optimisation were reviewed by Roy et al. [12]. PSO algorithms were used to solve discrete and continuous optimisation problems [13–16]. Rao et al. [17] reviewed the applications of advanced heuristic methods for design optimisation for mechanical components. Taktak et al. [18] optimised the helical spring design parameters using GA and many other heuristic methods.

Springs are vital elements for any suspension system used for restoring starting position after work done for the given load and as well as absorbing the strain energy to perform the work. In automotive, the spring supports the mass of the vehicle and as well as absorb the shocks due to the irregularities in the road conditions. Helical coil compression springs are commonly used as suspension system in a light automotive. Off late, many researchers have developed the optimal design of helical springs considering objectives separately. In this study helical spring is optimised with objectives namely minimizing volume and maximizing strain energy using multi-objective optimisation techniques considering a variety of realistic constraints.

1.2. Multi-objective optimization

Multi-objective Programming methods were developed to generate non-dominated solutions considering multiple objectives while satisfying physical limitations. Multi-objective Evolutionary Algorithms include techniques like Genetic Algorithm, Evolutionary Programming, Genetic Programming and Multi-Modal Function Optimization. Non-Elitist Multi-objective evolutionary algorithms include Vector Evaluated Genetic Algorithm, Random Weighted Genetic Algorithm, Non-Dominated Sorting Genetic Algorithm (NSGA), Niched-Pareto Genetic Algorithm, and Predator-Prey Evolutionary Strategy [19–22]. Multi-objective strategy using PSO approach also developed by the researchers [23,24] for solving continuous and discrete optimisation problems. Recently a review on multi-objective optimisation was presented by Gunantara [25].

It has been observed that in helical coil springs, volume and strain energy are two important parameters which have to be optimised. Volume objective minimizes the space requirements and weight. The weight intern reduces the cost of the spring. Strain energy increases the utility of the spring for a longer duration effectively. The multiobjective optimisation problem formulation considered in this study was taken from the study made by Sabarinath et al. [26]. They have solved the multi-objective optimisation problem using Adaptive Parameter Harmony Search Algorithm.

Finite Element Analysis (FEA) is a computer simulation technique that analyses the product design. FEA has become one of the standard practice to validate the product design. Taktak et al. [27] performed a dynamic FEA of helical spring. Stress distribution in helical coils is analyzed using FEA by Habuchi [28]. Sarkate [29] analysed helical coil springs made of high carbon steel. Lok-Choi and Ho-Choi [30] carried out numerical methods for composite helical springs. Pawar et al. [31] have carried out a static analysis on a three-wheeler front suspension coil spring. Dhanapal et al. [32] have analysed helical coil spring for a two-wheeler shock absorber. In this study, a helical coil spring made of IS 4454 used in light three-wheeler automotive is formulated as a multi-objective optimisation problem. The optimisation problem has been solved by considering two objectives a) Minimizing the volume of the spring and b) Maximizing the strain energy simultaneously using NSGA-II. NSGA - II will provide a set of non-dominated solutions. All the non-dominated solutions are unique, and the user can choose a suitable solution according to their requirement. FEA is carried out to validate the non-dominated solutions obtained by NSGA II.

2. Objectives, methodology and problem formulation

2.1. Objectives

The main objective of this study is to design a coil spring which is used in a commercial three-wheeler automotive using multi objective optimization procedure. The helical coil spring made of IS4454 is subjected to a constant axial compressive load. Objective functions considered in this study are a) Minimize the volume of the spring and b) Maximize the strain energy. The objective functions are subject to a set of constraints. The design variables are; Wire diameter (d), Mean coil diameter (D) and Number of active coils (N). The results obtained by the multi-objective optimization is validated using numerical analysis.

2.2. Methodology

The methodology adopted to meet the objectives of this study are summarized as follows:

Step1: Formulation of the optimisation problem: Defining Design Variable, Objective function, Constraints, and bounds

Step2: Solve spring design problem considering multiple objectives using classical multi-objective optimisation method (Weighted Sum Approach).

Step3: Solve spring design problem using Evolutionary Multi-Objective Optimization procedure: NSGA-II

Step4: Obtain non-dominated optimal solutions

Step5: Select 3 solutions from the non-dominated front (2 extreme solutions of the front and one from the middle of the non-dominated front)

Step6: Model the helical springs in “Solid Works Workbench” using the optimal parameters of helical spring obtained from the non-dominated front

Step7: Export the CAD models to “ANSYS workbench” and perform static FEA on the helical spring models and obtain the weight of the springs and strain energy

Step8: Compare the solution obtained from the Numerical model with the solutions obtained from the NSGA-II algorithm

2.3. Problem formulation

The multi-objective problem is formulated to find $\{\bar{X}\}$, which minimizes $f_1(X)$, and maximize $f_2(X)$. The objective functions are subject to constraints $g_j(X) \leq 0$, where $j = 1, 2, \dots, 8$. (In-Equality

Constraints). The objective functions $f_1(X)$, and $f_2(X)$ represents 'volume' and 'strain-energy' functions, respectively. The 'volume' function must be minimized and 'strain-energy' function has to be maximized. The design variables considered for optimization is shown in Equation (1).

$$\text{DesignVector}, \vec{X} = [d, D, N]^T \quad (1)$$

Where 'd' is the wire diameter, 'D' is the coil diameter and 'N' is the Number of coil turns. The 'volume' and 'strain-energy' objective function formulations are given in Equation (2) and Equation (3) respectively.

$$f_1(X) = \text{crosssectionalarea} \times \text{springlength} = \frac{\pi}{4} D d (N + 2) \quad (2)$$

$$f_2(X) = \text{strain} - \text{energy} = \frac{\pi}{4} f_{\max} \delta \quad (3)$$

Where, ' $f_{\max}$ ' is the maximum applied load and ' δ ' is the deflection of the spring and the deflection is given in Equation (4).

$$\delta = \frac{8 F D^3 N}{G d^4} \quad (4)$$

Where 'F' is the applied load, 'D' is the coil diameter, 'N' is the no. of turns, 'G' is the modulus of elasticity of the spring material and 'd' is the spring wire diameter. The various constraints imposed on the spring is given in Equations (5–12). Equation (5) represents the shear stress. The spring length under loading is given in the Equation (6). Wire diameter and coil diameter constraints are shown in Equation (7) and (8) respectively. Equation (9), (10) and (11) represents the preload condition and maximum load condition deflections, respectively. The spring index constraint is shown in Equation (12).

$$g_1(X), \text{ShearStress} = \frac{8 C K F_{\max}}{\pi^2} - S \leq 0 \quad (5)$$

$$g_2(X), \text{Springlengthunderloading} = l - l_{\max} \leq 0 \quad (6)$$

$$g_3(X), \text{Springlengthunderloading} = d_{\min} - d \leq 0 \quad (7)$$

$$g_4(X), \text{CoilDiameter} = (D - d) - d_{\max} \leq 0 \quad (8)$$

$$g_5(X), \text{Preload} - \text{deflection} = \delta_p - \delta_{pm} \leq 0 \quad (9)$$

$$g_6(X), \text{Max.deflectionLoading} = \delta_p \frac{(F_{\max} - F_p)}{k} + 1.05(N + 2)D - l \leq 0 \quad (10)$$

$$g_7(X), \text{Pre-load and Max.Loadingdeflection} = \delta_w \frac{(F_{\max} - F_p)}{k} \leq 0 \quad (11)$$

$$g_8(X), \text{Springindex} = 6.3 - C \leq 0 \quad (12)$$

The coil spring is designed using the IS4454 material. Material properties pertaining to IS4454 is considered for arriving the optimal values of wire diameter, coil diameter and number of coils. Typical chemical composition of spring steel material, IS4454 used for making automotive springs consists of C – 0.50 to 0.60, Si – 0.10 to 0.35, Mn – 0.50 to 0.80, S – 0.05 (max.) and P – 0.05 (max.). The important parameters of the helical coil spring and their values are as shown in Table 1.

Table 1

Parameters used in the spring design problem.

Parameter name	Abbreviation	Value
Max. load	$F_{\max}$	3200 N
Max. shear-stress (allowable)	S	1303 MPa
Max. Free length	$l_{\max}$	315 mm
Min. wire-diameter	$d_{\min}$	10 mm
Max. outside spring diameter	$D_{\max}$	88 mm
Compression force (preload)	F_p	1334 N
Max. allowable deflection	δ_{pm}	152 mm
Deflection	δ_w	32 mm
Shear-Modulus	G	81370 MPa

3. Solution methodology using weighted sum approach

3.1. Problem formulation

Weighted sum approach is a traditional technique used to solve problems with multiple objectives by assigning weights to objective functions. A typical problem consisting of 'n' no. of objective functions are shown in Equ. (13). Objective functions are formulated as a single objective function using appropriate weights, and heuristic approaches are used to arrive the solution.

$$\text{Minimizef}(\vec{X}) = w_1 f_1(X) + w_2 f_2(X) + \dots + w_n f_n(X) \quad (13)$$

Subject to constraints as given in Equations (5–12)

Where w_1 and w_2 are the weights assigned to objective functions such a way that $w_1 + w_2 + \dots + w_n = 1$. Weights are the relative importance of the objective functions. The user can choose weights depends on the application or priority. For example, if there are two objectives say, minimizing volume and maximizing strain energy. If the user is interested to get a trade-off solution considering equal importance, the user can choose the weight, w_1 for volume as 0.5 and, w_2 for strain energy as 0.5. Weighted sum approach does not guarantee non-dominated solutions but provides a variety of solutions for the user by varying the weights of the objective functions. The combined objective function is shown in Equ. (14). The constraints are given in the Equ. (5) to Equ. (12).

$$\begin{aligned} \text{Minimizef}(\vec{X}) &= w_1 f_1(X) + w_2 f_2(X) \\ &= w_1 \left[\frac{\pi}{4} D d (N + 2) \right] + w_2 \left[-\frac{\pi}{4} F_{\max} \delta \right] \end{aligned} \quad (14)$$

Weights assigned to objective functions $f_1(X)$ and $f_2(X)$ are w_1 and w_2 respectively. Here f_1 is the volume function, and f_2 is the Strain Energy function. Strain energy function must be converted into minimization form while writing the objective function.

3.2. GA implementation

Experiments were carried out using a GA. Objective functions are assigned with different weights as shown in Table 2.

GA is initialized with a population size of 150. Tournament selection with single-point crossover is used to generate the intermediate population. Cross over, and mutation is carried out with a probability of 0.80 and 0.05 respectively. Maximum no. of generation allowed for GA for termination is 10000. The results produced by GA for the different weights assigned for volume and strain energy functions are shown in Table 3.

Table 2

Weights used for objective functions.

w_1 (Volume)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
w_2 (Strain Energy)	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0

Table 3

Results obtained by GA for weighted sum formulation.

w_1	w_2	d	D	N	Combined Fitness
0.0	1.0	12.02	75.86	14.95	6.18E-06
0.1	0.9	12.01	64.48	13.05	34444.57
0.2	0.8	12.00	64.56	13.00	68747.77
0.3	0.7	12.00	64.56	13.00	103120.04
0.4	0.6	12.00	64.56	13.00	137493.52
0.5	0.5	12.00	64.50	13.03	172115.80
0.6	0.4	12.00	64.48	13.05	206632.39
0.7	0.3	12.00	64.56	13.00	240613.73
0.8	0.2	12.00	64.47	13.05	275580.80
0.9	0.1	12.00	64.48	13.05	309981.81
1.0	0.0	12.00	64.55	13.00	343800.78

4. Multi-objective optimization of helical spring design problem using NSGA-II algorithm

4.1. NSGA – II framework

NSGA-II is a heuristic algorithm to find non-dominated solutions or Pareto fronts of multi-objective optimisation problems. This algorithm was developed as its previous version NSGA faced problems like high computational complexity, lack of elitism, and need for specifying sharing parameter as highlighted in Deb et al. [20]. The features of NSGA-II include elitism, which means storing of all non-dominated solutions enhancing faster convergence properties and evaluation of crowding distance which ensures diversity and spread of the solutions. The working of the NSGA-II algorithm is illustrated in Fig. 1.

The objective function and constraints of the multi-objective optimisation problem are given in the Equation (1) to (12). The constrained problem is converted into an unconstrained problem using penalty function approach. A bracket operator penalty term is used to tackle the constant violations. The fitness function used in this study is shown in the Equation (15).

$$\text{FitnessFunction}, P(X) = f(X) + \sum_{j=1}^J R_j \langle g_j(X) \rangle^2 \quad (15)$$

where, $f(X)$ is the objective function, ' R ' is the penalty parameter, $g_j(X)$ is the constraint function and ' J ' is the maximum number of constraints. Simulations were carryout with a population size of 150, tournament selection operator, single point crossover with a probability of 0.80, mutation probability of 0.05, and pareto front population fraction as 0.35. Simulations carryout with a termination criterion considering maximum number of generations as 10000. Details are shown in Table 4.

4.2. NSGA – II implementation

The NSGA-II is implemented to obtain non-dominated solutions for spring design multi-objective optimisation problem. The NSGA-II generated a total of 18 solutions for multi-objective optimisation problem considered in this study. Each solution is unique with different objective function values for 'Volume' as well as 'Strain Energy'. The corresponding Pareto front solutions are given in

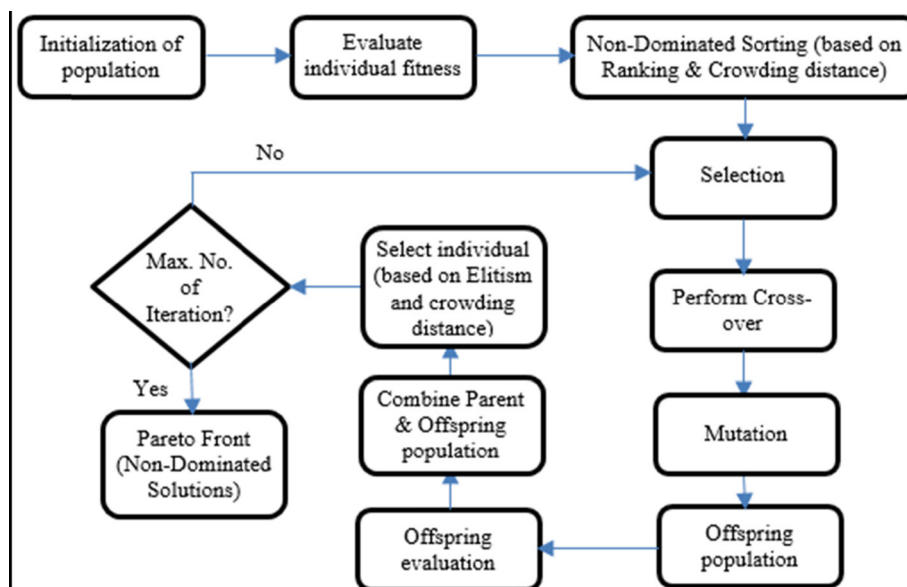
**Fig. 1.** Working of NSGA-II (Deb et al. 2002).

Table 4

Parameters used in NSGA-II for convergence.

NSGA II parameter	Value
Population size	150
Selection operator	Tournament selection
Cross over type and probability	Single point, 0.80
Mutation function	Adaptive feasible
Mutation probability	0.05
Distance-measure	Crowding-Distance
Non dominated front pop. fraction	0.35
Max. No. of iterations	10,000
Function Tolerance	1e-10000
Constraint Tolerance	1e-10000

Table 5. Each solution is unique, and the designer can interpret and choose the result which confirm to their requirements.

5. Finite element modelling and analysis of helical coil spring

5.1. Selection of helical coil springs for FEA

The solution obtained from the Pareto front is validated using FEA. For numerical modelling, three spring designs were taken from non-dominated solutions provided by NSGA II. Two extreme designs of the Pareto front and one from the middle of the front are chosen for analysis. The details of the helical springs selected for the analysis is shown in [Table 6](#). The primary purpose of this analysis is to measure the strain energy stored in the spring for the maximum loading condition to confirm the solution provided by NSGA II.

5.2. FEA modelling of helical coil spring

The helical coil springs were modelled using 'Sledworks' software. The models are imported into 'ANSYS workbench' in "IGES" format. The material used for the spring is 'IS 4454'. The modulus of elasticity of the material is 20600 MPa, density is 7.85 g/cm³, Poisson ratio is 0.3, Ultimate tensile strength is 1250 MPa and Modulus of rigidity is 81370 MPa. After assigning the material properties, the models are meshed. No. of nodes used for modelling are 331,558 and No. of elements considered are 184818. The Boundary conditions were applied in the form of fixed support on one end and force on the other end. A load of 3200 N is applied over the helical coil spring to study the objectives considered in this study.

Table 5

NSGA-II results.

Index	Volume(mm ³)	Strain Energy,(Joules)	dWire-diameter,(mm)	DMean coil-diameter,(mm)	NNo. of active coils(Nos.)
1	228742.70	327046.31	10.00	64.00	12.50
2	261081.80	405378.12	10.00	66.19	14.00
3	251910.80	385563.12	10.00	65.87	13.51
4	242262.00	371982.43	10.00	66.20	12.85
5	239379.80	366788.20	10.00	66.19	12.67
6	253157.70	391316.92	10.00	66.20	13.51
7	255811.80	392417.43	10.00	65.87	13.75
8	234211.10	336115.64	10.00	64.00	12.85
9	247259.90	377225.71	10.00	65.86	13.23
10	239408.00	366918.09	10.00	66.20	12.67
11	242262.00	371982.43	10.00	66.20	12.85
12	231548.70	331570.29	10.01	64.01	12.68
13	235440.30	356623.39	10.00	65.87	12.50
14	248542.80	383127.67	10.00	66.20	13.23
15	228742.70	327046.31	10.00	64.00	12.50
16	231451.90	331539.60	10.00	64.00	12.67
17	236605.70	361945.32	10.00	66.20	12.50
18	261081.80	405378.12	10.00	66.20	14.00

Table 6

Design variables of the springs selected for modelling.

Model	Dia. of wire, (d)(mm)	Mean dia. of Coil (D),mm	No. of coils,(N)
Spring1	10.00	64.00	12.50
Spring2	10.00	66.20	12.85
Spring3	10.00	66.20	14.00

5.3. FEA results

FEA analysis is carried out to study the static behaviour of the coil spring. Strain energy is computed for all the springs. Helical spring under loading conditions with obtained strain energy is shown in [Figs. 2, 3 and 4](#) for three different types of spring configurations taken from Pareto front obtained using NSGA- II. The strain energy and volume obtained for each of the spring configurations are shown in [Table 7](#).

6. Results and discussions

In this study, Multi-objective optimisation of helical coil spring of a light three-wheeler has been carried out using weighted sum approach and NSGA II. Weighted sum approach is a simple technique, and solutions can be obtained with ease by assigning weights to the objective functions. But main disadvantage in weighted sum approach is that the solution obtained are not non-dominated solutions. NSGA II is a popular multi-objective optimisation tool for generating Non-dominated solutions. Set of Pareto optimal solutions were obtained. Each solution is unique and is non-dominated. Users can choose any of the 18 non-dominated solutions obtained by NSGA II. The extreme solutions in the Pareto front indicate the optimum solutions of the problem considering objective functions separately as a single objective one. From the Pareto front, extreme solutions and solution from the middle are chosen for validation using FEA. Strain Energy and Volume of the springs were calculated and compared with the solution obtained by NSGA II. Relative % error in strain energy between NSGA II and FEA is also computed. The comparison is shown in [Table 8](#). The results provided by FEA for the design of the spring provided by NSGA II are very close. This means NSGA II spring designs are safe and could be used for automotive application. The manufacturer can choose a suitable spring design for

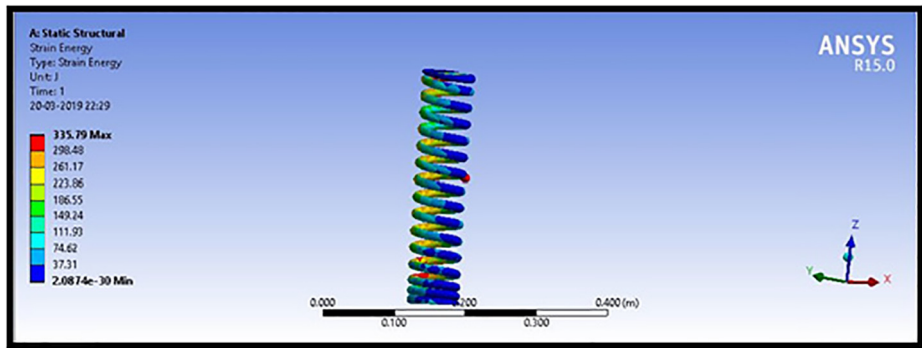


Fig. 2. FEA of Spring 1.

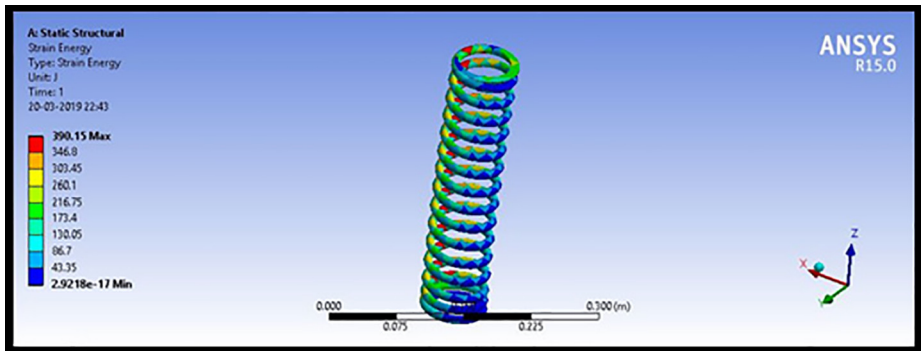


Fig. 3. FEA of Spring 2.

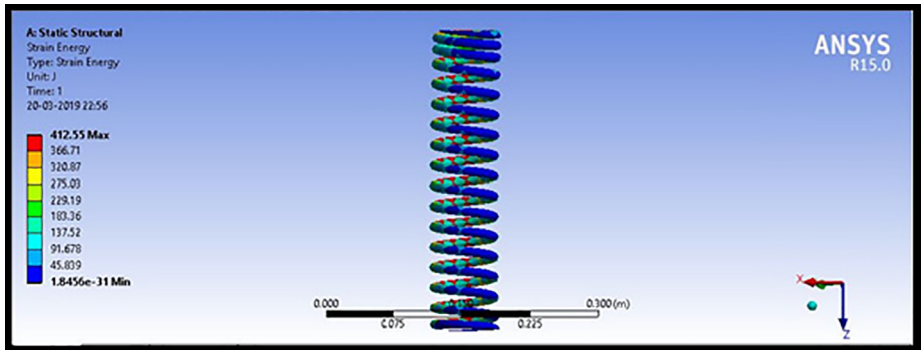


Fig. 4. FEA of Spring 3.

Table 7
Strain energy and volume of the helical springs.

Model	Strain-Energy, Joules	Volume, mm3
Spring 1 (d = 10 mm, D = 64 mm, N = 12.5)	228742.70	327046.31
Spring 2 (d = 10 mm, D = 66.20 mm, N = 12.85)	242262.00	371982.43
Spring 3 (d = 10 mm, D = 66.20 mm, N = 14)	261081.80	405378.12

the automotive based on their priority from the pareto solutions generated by NSGA II.

7. Conclusions

In this paper, a helical coil spring made of IS 4454 used in the automotive application has been optimised using multi-objective

optimisation procedure. Bi-objectives namely, minimizing the volume and maximizing the strain energy were considered in this study along with realistic constraints. Non-dominated pareto solutions were generated using NSGA II. A total of 18 non-dominated solutions were generated by NSGA-II. To validate the solutions generated from NSGA II, FEA has been carried out by selecting extreme solutions of the Pareto front and middle one. Strain energy and volume of the springs were computed using the FEA procedure, and the results were compared with the pareto solutions generated by NSGA II. It is observed that there is a close correlation exists between the solution obtained by NSGA II and numerical procedure. A maximum relative percentage error of 4.88% was observed with respected to strain energy. The results show the effectiveness of NSGA II. The designer can choose any one of the non-dominated solution obtained by NSGA II for the automotive application based on their priority.

Table 8
FEA results comparison.

Spring Model	NSGA II Results		FEA results		Relative % error(Strain Energy)
	Volume (mm ³)	Strain energy (Joules)	Volume (mm ³)	Strain energy (Joules)	
1	228742.70	327046.31	228742.70	335790.00	2.67
2	242262.00	371982.43	242262.00	390150.00	4.88
3	261081.80	405378.12	261081.80	412550.00	1.77

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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